Capstone Team 76 Progress

User Objectives:

- Research Paper Recommendation for Individuals
  - Based on the user's experience level and interests, we will recommend required articles and research papers

- Research Paper Recommendation for Groups
  - Based on the group's experience level and interest, we will recommend required articles and research papers

- Problem Statement Generation
  - For a user/group, internally we will get the research papers
  - Each paper has *keywords* , we will use this to query an LLM
    - LLM like Ollama

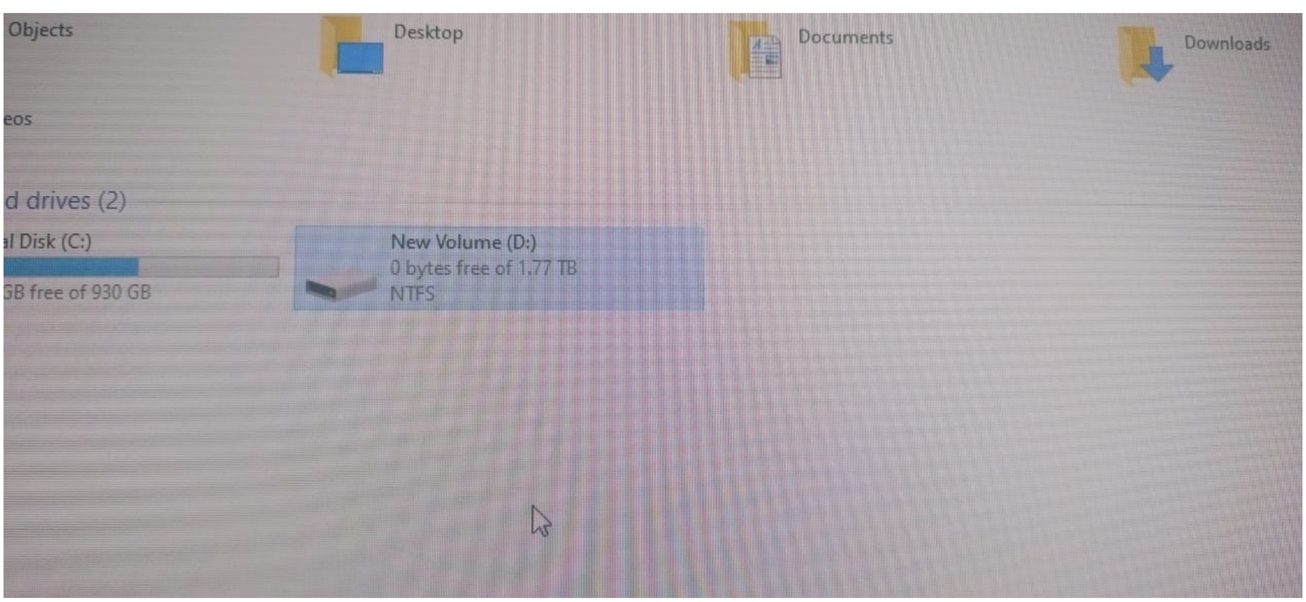
  - Or we can take into consideration the under explored connections between topic entities and query an LLM

- Researchers Recommendation
  - Based on user's institutions, and interests, we can recommend researchers they would like to follow or keep track of

User Objective	Algorithm to Use	Model/Approach
1. Research Paper Recommendation for Individuals	- Content-Based Filtering (CBF)	- TF-IDF, Doc2Vec (we considered SciBert, Bert but they require multi-gpu and we don't have those resources.)
	- Collaborative Filtering (CF)	- Matrix Factorization (good for handling large data)
	- Knowledge Graph-Based Retrieval	- Personalized PageRank for user-paper relevance ranking
2. Research Paper Recommendation for Groups	- Hybrid Filtering (CBF + CF)	- Weighted hybrid model (CBF + CF) to balance content and user preferences
	- Grey Wolf Memetic Algorithm (GWMA)	- Optimization algorithm for group preference aggregation
	- Group Consensus Algorithms	- Preference aggregation (Least Misery, Average Satisfaction)
3. Problem Statement Generation	- Knowledge Graph Exploration	- Graph-based underexplored topic connections
	- LLM-Based Querying	- Ollama (Phi, Mistral, Llama3) to generate problem statements
4. Researchers Recommendation	- Content-Based Filtering (CBF)	- TF-IDF/BERT embeddings to match researchers' papers with users' interests
	- Social Graph Analysis	- Personalized PageRank on co-authorship & citation network
	- Institution-Based Clustering	- K-Means, DBSCAN for finding researchers from similar institutions
	- Collaborative Filtering	- User-based CF to recommend researchers followed by similar users

Challenges Faced :

- Storage Issues
  - D Drive got full
  - We have a couple of zipped folder due to which we haven't been able to finish 100% preprocessing
  - the scripts are ready and working its just that there's no *space*.



Update:
We are transferring the data not useful for the project in the D Drive to an external hard disk and are planning to distribute it in the other systems - it's taking some time.

RoadMap

- Data Preprocessing. - 90% done
  - Data Reduction - done
  - Feature Extraction - done
  - a couple of folders left which we are struggling to do on account of there being no space. We are finding a solution for this issue though.
- Build Knowledge Graph of the research network
- Make embeddings for nodes in the graph
- CBF-CF hybrid algorithms
- Grey Wolf Algo , Memetic Algo
- Group preference modelling
- Evaluation & Tuning

What are we asking the users:

- experience level - (novice, intermediate, advanced)
- interested domains - (*map to topics/concepts/domains*)
- which institution ?
- author or no?
- do you want problem statement or just papers ?
- are you part of a group ?
  - if yes, give their id's for our website
    - OR
      - give all this info(experience_level, interested_domains, etc) for everyone

Outcomes

- Knowledge Graph creation with network between topics, concepts, authors and works.
- Research Paper and Authors recommendation for Individuals
- Research Paper and Authors recommendation for Groups
- Problem Statement Generation

Dataset

- Our dataset has 10 entities:
    - Authors
    - Concepts
    - Domains
    - Fields
    - Institutions
    - Publishers
    - Sources
    - Subfields
    - Topics
    - Works

Each entity has multiple json files.
We have attached one entry from the json file to the drive.

https://drive.google.com/drive/folders/1r4B28MN9nG79iqk6Plsrj9UPGxLfmqzv?usp=sharing